VoiceOps 2026 — ZentroMart voice support agent

Author: Shreshtha · **LLM:** Gemini `gemini-3.1-flash-lite` (free tier, OpenAI-compatible API) · **Speech:** Gnani `gnani-vachana` 0.8.2 (Prisma STT, Timbre v2.5 TTS) · **Code:** `run.py`, `src/` (4 modules), `eval/`, `tests/`

1. What we built, step by step

Step 1: prompt only. One LLM call. The system prompt sets the role (ZentroMart's AI support assistant), the limits (no promised refunds or actions; say when unsure) and the length. No documents.

Step 2: grounded in the documents.

- **Chunking:** each `## [SECTION-ID]` heading is one chunk: 84 chunks of 6–135 words, so a rule is never cut off from its heading.
- **Metadata:** every chunk keeps its `doc_type`, `status` and `effective_from`.
- **Retrieval:** BM25 (`bm25s`, with a romanized-Hindi stoplist) and multilingual embeddings (`intfloat/multilingual-e5-small`, numpy index), fused by reciprocal rank.
- **Query rewrite:** before retrieval, an LLM call rewrites the message into a standalone English query. The raw message also counts, at half weight.
- **Answering:** the model sees the top 6 sections sorted and labelled by authority (current policy > agent guide > help > FAQ > marketing; superseded sections marked). It must return the section IDs it used.

Step 3: a real conversation. Code, not the model, owns everything that must not go wrong.

- **Verification state machine.**
  - An order ID triggers `lookup_order(order_id)` and a request for the phone's last 4 digits or the delivery pincode.
  - The answer triggers the verified lookup. One retry per order, then refusal (IDV-3); naming another order does not reset it.
  - Phone, pincode and customer ID are stripped from tool results, so **private data never reaches the LLM**.
- **Policy maths in Python** (`src/rules.py`, unit-tested against the practice gold):
  - Business days count from the day *after* initiation, Mon–Sat, skipping the holiday table. This reproduces all 9 dated refund answers.
  - Return windows are calendar days, with the RET-3 exception (pre-August electronics: 10 days) and RET-8 (festive fashion: 30 days).
  - Refund arrival ranges per payment method, and the "more than 3 business days overdue" check.
  - The ₹49 pickup fee, refund timing after refusing at the door, and the refund reference number.
- **Policy-owned decisions.**
  - The ESC-1 triggers force an escalation: human requested, fraud, legal, safety, ≥2 earlier contacts, item > ₹10,000, or an overdue refund. The model cannot escalate on its own.
  - A compensation demand is refused unless it is the one case DEL-4 pays for (a late Express order).
- **Guards.**
  - A ticket is only mentioned after `create_ticket` returns `created=True`.
  - Replies claiming a cancellation, approval or credit, or showing private digits, are regenerated.
  - Evidence IDs that were not shown to the model are dropped.
  - **Abstention gate:** the model must mark whether the evidence actually answers the question. If not, code replaces the reply with "I don't have that information" plus an offer of a human colleague.
  - **Must-state post-check:** where a policy requires a computed fact (the refund reference number when a verified customer asks about a refund, REF-6; the 7–10 day refund timing when a shipped order's cancellation is refused, CAN-2), code checks that the reply contains its numbers. If not, one rewrite is requested. It is kept only if it states the fact, keeps every number of the first reply and passes the other guards; on any failure the first reply stands.

Step 4: voice.

- **Input:** `run.py` converts any audio format to 16 kHz mono, splits it into ≤30 s pieces and transcribes with Gnani `en-IN`.
- **Hindi second pass:** if the transcript looks Hindi (Hindi function words or Hindi number words), a second `hi-IN` pass runs and the agent receives both transcripts. `en-IN` writes spoken Hindi numbers as words ("saat hazaar…"); `hi-IN` returns digits.

- **Speech output:** replies are written for the ear (dates with month and year, order IDs in groups of four) and spoken with Kaveri (en-IN), Bhavna (hi-IN) or Poorvi (Hinglish, via the documented `hi-en` REST code, which the SDK rejects).
- **Robustness:** silence gets "Sorry, I didn't catch that". Sub-folders of audio are treated as one conversation. A failing file never stops the batch.

2. Evaluation

How it is scored. All numbers are on the dataset's practice files and come from `eval/evaluate.py`. A reply is **fully correct** only if all of these hold:

- the decision is one of the acceptable ones
- every `must_include` fact is present (dates matched by value across English and Hindi formats, numbers as whole numbers)
- no `must_not_include` phrase appears

Gold-evidence recall is reported separately. Result files are in `eval/results/<model>/`.

Step 1 vs Step 2 vs full agent: 52 single questions (gemini-3.1-flash-lite)

	Step 1: prompt only	Step 2: + RAG	Full agent	Full agent, noisy `asr_text`
Fully correct	15%	83%	**96%**	**96%**
Required facts present	23%	87%	98%	98%
Acceptable decision	75%	90%	98%	98%
Forbidden phrase used	0%	0%	0%	0%
Gold-evidence recall	0.00	0.72	0.83	0.85
Fully correct: English / Hindi / Hinglish	18 / 12 / 9%	82 / 88 / 82%	97 / 100 / 91%	97 / 88 / 100%

What the jumps show.

- **Step 1 → Step 2:** retrieval is worth 68 points. Without documents the model answers fluently and wrongly, for example the generic "5–7 days" for every refund.
- **Step 2 → full agent:** worth another 13 points, from:
  - catalog rows (warranty months, the Assured rule for marketplace sellers)
  - the date calculators
  - escalation decided by policy rules
  - the abstention gate
- **Noisy transcripts:** speech-recognition text (`asr_text`: spelled-out digits, misheard words) costs nothing overall, thanks to the transcript normaliser and the English rewrite.

Retrieval (section IDs, 46 answerable questions):

Query used	R@1	R@3	R@5	MRR	R@5 Hindi
Raw customer text	0.30	0.48	0.68	0.50	0.43
English rewrite + raw at half weight	0.43	0.78	**0.90**	0.69	**0.86**
Same, on noisy `asr_text`	0.45	0.74	0.85	0.69	—

Conversations: 23 conversations, 96 customer turns.

- 96% of turns fully correct, **100% correct decisions, 100% exactly the expected tool calls** (`lookup_order` with and without the verification value, `create_ticket`), evidence recall 0.92.
- By split: dev 72/75, held-out test 20/21. The must-state post-check was built on dev failures; no test turn triggers it, so test is unchanged (20/21 before and after).
- Per scenario:
  - **100%:** verification failure, high-value damage, human requested, repeat contact, return eligibility, topic switch, Express late credit, cancelling an unshipped or a shipped order
  - **88%:** compensation demand, refund status
  - **75%:** where-is-my-order

Voice end to end. A 17-file test set, made with Gnani TTS plus a silent clip, ran through `run.py` with the batch never stopping: 8 questions in three languages, two 4-turn conversations (Hinglish high-value damage, Hindi failed verification) and silence.

- In the Hindi conversation, the caller's spoken code was read correctly: the `hi-IN` pass gives "1,266", which the extractor turns into `1266`. The agent then made the gold tool calls: verify, ask again, refuse.
- **Latency.** On the first live run a voice turn took 7–50 s on the free tier, much of it rate-limit waiting and retries. A rerun with the LLM answers cached, so speech only, took a median of 6.5 s (3–20 s): that is the Gnani STT + TTS share of a turn.

Model comparison (same code; `gemini-3.5-flash-lite` was used first):

- Fully correct: 21% (Step 1), 92% (full agent).
- Its Step 2 figure, 67%, was measured before the prompt fixes. Its conversation run was cut short by the daily quota.
- 3.1 was also faster in our probe: 2.3 s vs 5.2 s per call.

3. What went wrong, and what we fixed

Problem the evaluation found	Fix
The section parser captured one section per file (`.*` under DOTALL swallowed the rest)	Anchored the heading pattern; a test asserts 84 sections
Dates in the query ("17 October 2026") pulled in sections that merely mention October	Stripping dates helped English but hurt festive-sale questions, where the date *is* the topic → reverted; the English rewrite solved it
The model invented answers to unanswerable questions ("GST invoice mil sakta hai")	`covered` flag + abstention gate
The model escalated on its own, and treated an Express-late question as compensation	Escalation comes only from the ESC-1 rules; a DEL-4 rule for compensation
Follow-ups ("and what if I'm not at home?"): the raw follow-up outvoted the context-aware rewrite	The raw query counts at half weight
Spoken Hindi numbers: `en-IN` gave "saat hazaar…", and Gnani's `hi-IN` formatting wrote "7,746 1,439"	Hindi number words trigger the `hi-IN` pass; the extractor drops thousands separators and joins split digit groups
A Hindi caller reading out digits was answered in Hinglish	The conversation keeps its language on such turns
In 3 conversation turns the decision was right but a computed fact was left out (refund timing after refusing a parcel, the refund reference number)	Must-state post-check: a rewrite only for those turns (3 extra LLM calls in the whole evaluation); conversations 93% → 96%
Our own review found that naming a different order reset the verification retry, allowing unlimited guesses	Failed attempts are counted per order for the whole conversation
Gnani's reply WAVs carry placeholder sizes (`0xFFFFFFFF`), so strict readers saw ~24 h of audio	The file is rewritten once with real sizes after synthesis
The free LLM tier ran out mid-evaluation (~550 calls/day per model)	On-disk response cache; failed rows refilled on a later run
Tried down-weighting superseded sections in retrieval	Rejected: it dropped the gold `RETv3-2` for the pre-August electronics case the policy still covers

Still open, and what we would fix next.

- **Too cautious on "what if I'm not at home?"** (2 turns). The abstention gate is conservative there. Next step: tune the rewrite toward the documents' wording ("delivery attempt").
- **Two single-question failures.** One unanswerable question (store pickup) still got an invented answer the gate didn't catch, and one asked an unnecessary "which colour?" before answering.
- **Correct but differently worded.** 2 failures use a Hindi synonym the strict string scorer doesn't list ("Mujhe khed hai" for "sorry", "क्वालिटी चेक" for "quality check"). A semantic scorer would pass them.
- **Spoken SKUs.** "ELE zero zero one two" comes back from STT as "s q le 12" and can't be recovered. Gnani's word-boosting (`bias_list`, REST only) is the next step.

4. Bonus work

- **Tests:** 55 automated tests, including scripted-LLM conversation tests (no API calls) for verification, ticket rules, abstention, the must-state post-check and language.
- **Evaluation tooling:** a Gnani-TTS test-audio generator; results stored per model; retrieval scored with and without the rewrite.
- **Live demo:** a Gradio app (`app.py`) showing each turn's decision, cited sections, tool calls and verification state.